

Investigating Turbulent Parameters In Forward Freestream Conditions For a Ducted Fan Inlet Using Computational Fluid Dynamics

AE 4040 Final Project

Jonathan Yun
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Executive Summary

During static testing of a 70 *mm* 3D-printed ducted fan, the fan inlet (a curved lip that expands the frontal section of the fan to a larger diameter) was identified as one of the most impactful design features. An inlet was experimentally constructed to maximize thrust. However, the static nature of this test does not account for the effects of forward freestream velocity, particularly those related to turbulence. Simulations were conducted to analyze pressure, velocity, turbulent kinetic energy, and wall shear stress distributions across three freestream velocities, corresponding to distinct turbulence conditions: low-power/start-up, cruise/efficient flight envelope, and high-speed/max-throttle scenarios. For low- and medium-turbulence simulations, only minor visual changes were observed in the contour and streamline plots. In contrast, the high-turbulence simulation revealed significant differences in pressure differentials, velocity magnitude, and overall turbulent kinetic energy and wall shear stress. Based on these results, the inlet appears to perform most efficiently in highly turbulent conditions, as indicated by enhanced air acceleration and favorable turbulent patterns. However, this report acknowledges its limitations in the number of simulations conducted and the scope of turbulence analysis.

1. Introduction & Motivation

Electric ducted fans (EDFs) are increasingly valuable for vertical takeoff/landing and light-lift aerospace applications due to their modularity, efficiency, and favorable acoustic signature. The design of the fan inlet plays a critical role in the EDF's performance; its shape and size significantly influence the mass flow of air delivered to the fan blades, positioned just beyond the inlet section. Using a custom EDF, a 3D-printed inlet lip, and a lever-based testing setup to validate thrust data, an inlet was designed to optimize thrust for an 11-blade, 70 *mm* ducted fan.

However, this testing setup had inherent limitations. The static testing did not account for positive freestream velocity, which directly affects the Reynolds and Mach numbers of the incoming flow and, consequently, the EDF's performance. Further testing and simulation are needed to investigate turbulence, boundary layer formation, and other flow characteristics influenced by forward freestream velocity. Computational fluid dynamics (CFD) simulations are preferred over mechanically complex methods, such as wind tunnel, hover, or in-flight testing, and are used here to characterize the EDF's performance across various freestream velocities. CFD also allows for a more comprehensive analysis of key flow properties, including pressure, velocity, and turbulence (e.g., turbulent kinetic energy and wall shear stress), enabling conclusions with greater precision and nuance.

2. Problem Design & Formulation

The 2D nature of this simulation necessitates an upper and lower wall, symmetrical about the x-axis, resulting in a 4 walled grid (inlet, outlet, and two walls). Most flow parameters are expected to remain consistent through the midsection of the inlet where the flow in itself is mostly non-turbulent given standard operating conditions of a 70 *mm* EDF.

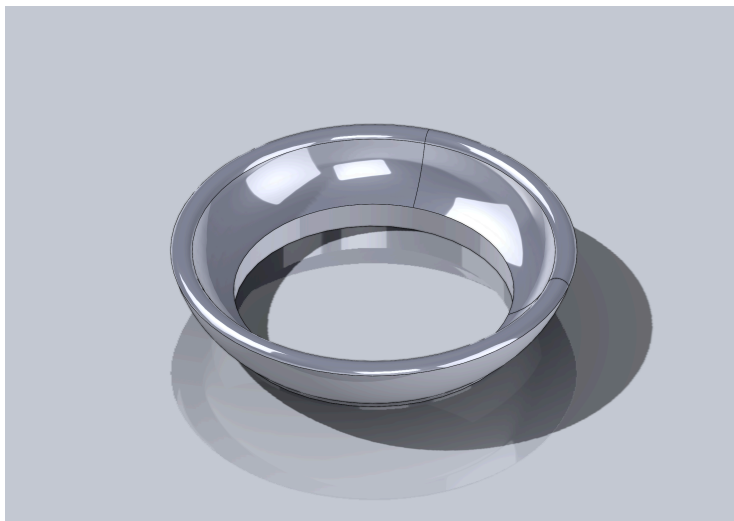


Fig 1. SOLIDWORKS render of the 70 *mm* EDF inlet to be 3D printed

Turbulence likely occurs at the viscous boundary layer formed at the walls and around the area reduction curves (i.e. sharp turns) at around $\frac{l}{2}$ of the inlet length. Therefore, sound grid construction at the boundary layer in particular is critical. Grid (or mesh) construction is done in Pointwise. Wall spacing is calculated using the Cadence y^+ calculator [1], which takes inputs as justified in Table 1.

Flow Parameter	Value	Justification
Freestream velocity	$100 \frac{m}{s}$	EDFs generally operate in subsonic, incompressible conditions. Mach 0.3, or $100 \frac{m}{s}$ at sea level, is used as the upper bound of velocity (and thus turbulence) parameters
Freestream density	$1.225 \frac{kg}{m^3}$	Freestream density at sea level
Dynamic viscosity	$0.000017894 \frac{kg}{m-s}$	Assumed dynamic viscosity of air
Reference length	$0.08 m$	Diameter of inlet (boundary condition)
y^+	1	Cadence recommended y^+ value [2]

Table 1. Flow parameter justifications for wall spacing calculations

This yields a wall spacing of $0.0002838 m$. Wall spacing is applied symmetrically to both upper and lower walls and a visible boundary layer is formed. With 92 points applied to every boundary, the grid is fully constructed and shown below.

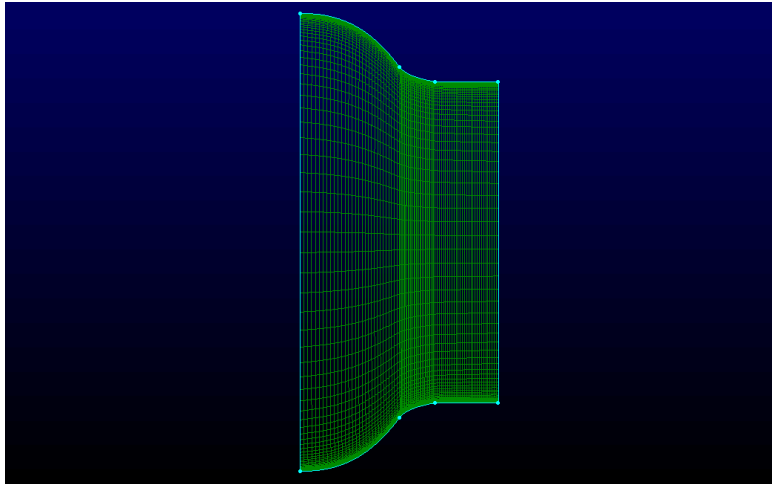


Fig 2. Final grid (or mesh) constructed in Pointwise

This grid has the following dimensions, as measured by Pointwise.

Dimension	Value
Cell count	8190
Area (min/max) (cm^2)	2.622e-6/0.0099
Skewness (min/max)	2.667e-5/0.6846
Aspect Ratio (min/max)	1.0003/142.77

Table 2. Grid dimensions for final mesh

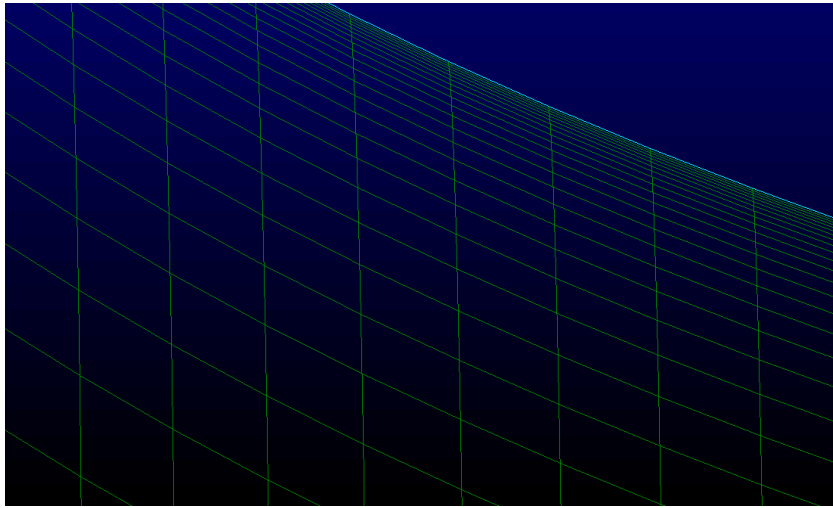


Fig 3. Zoom-in on boundary layer mesh

The flow parameters specifically related to the positive freestream velocity revolve around velocity and turbulence quantities. Thus, Reynolds numbers are used to demarcate flow conditions, and the corresponding velocities are used to construct the boundary conditions.

All flow conditions considered in this report are incompressible and the simulations are assumed to operate at the same altitude. With the Reynolds number equation solving for velocity,

$$u = \frac{Re\mu}{\rho l} \quad (\text{Eqn. 1})$$

all variables besides Re and u are constant. Reynolds numbers of 3,500, 50,000, and 200,000 are used to model transitioning turbulent flow on startup, a realistically efficient flight envelope, and high speed turbulent conditions respectively. This corresponds to velocities of 2.44, 12.21, 48.86 $\frac{m}{s}$ which is applied at the inlet wall.

Four turbulent related quantities are plotted in this report to comprehensively compare the three Reynolds scenarios. The quantities are justified in Table 3.

Turbulent Quantity	Justification
Static Pressure	One of the most effective indicators of fan performance in general [3]. Turbulence has a direct negative effect on static pressure because of its chaotic nature (dissipating energy and separating flow), so low static pressure regions can be attributed to high turbulence.
Velocity	Turbulence is readily visualized using velocity geometry. Swirls, eddies, and separation are all clear indicators of turbulence [4]. A streamline plot of velocity is used for the most direct study of velocity direction.
Turbulent Kinetic Energy	Turbulence kinetic energy (TKE) is “directly related to the transport of momentum and heat through the boundary layer, and it indicates an energy limit that stimulates the creation and dissipation rates of eddies” [5]. Comparison of TKE between simulations is a direct comparison of turbulence.
Wall Shear Stress	Wall shear stress is a direct result of the presence of the turbulent boundary layer at the walls. A visualization of the walls will highlight wall regions of high turbulence, which excites the shear wave as a consequence of the no-slip condition at the walls [6].

Table 3. Turbulent quantities to be plotted and their contribution to the conclusion

3. Results & Discussion

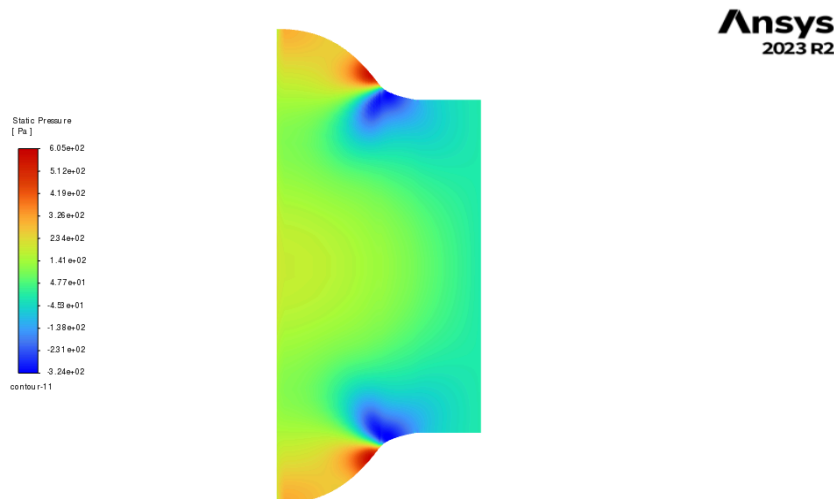


Fig 4. Static pressure contour plot at $2.44 \frac{m}{s}$ inlet velocity

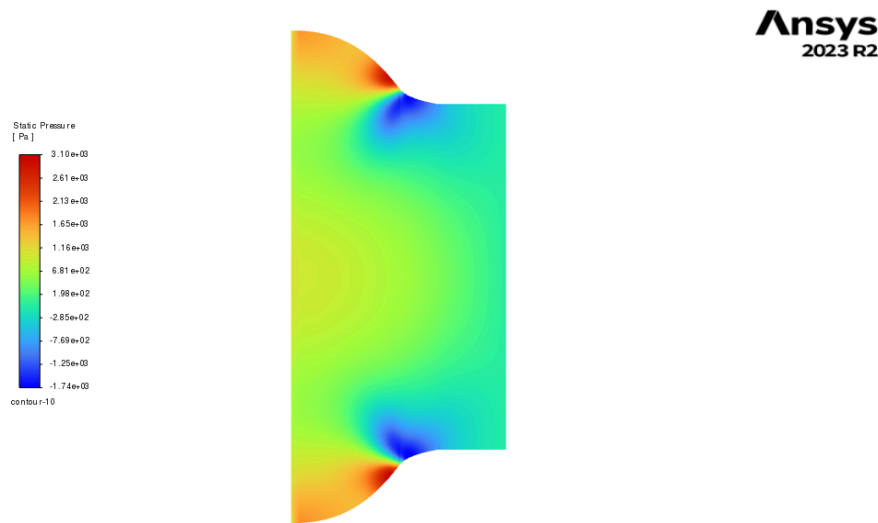


Fig 5. Static pressure contour plot at $12.21 \frac{m}{s}$ inlet velocity

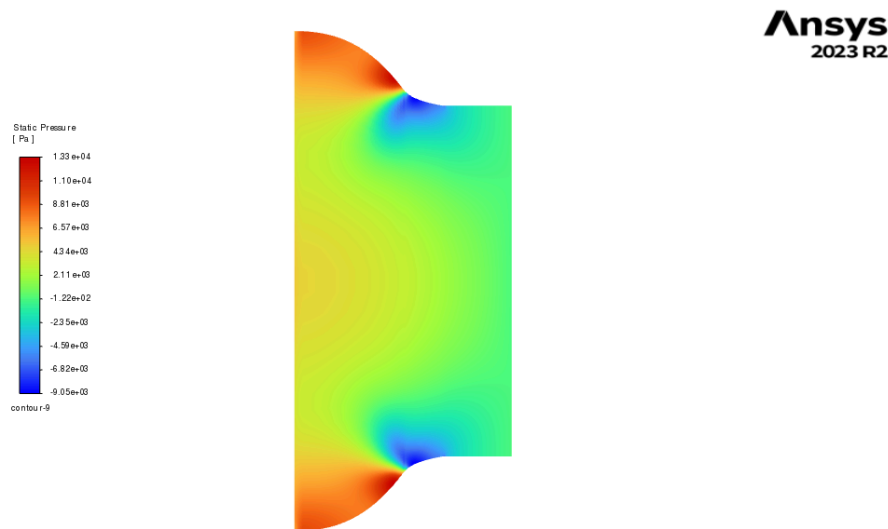


Fig 6. Static pressure contour plot at $48.86 \frac{m}{s}$ inlet velocity

At low ($2.44 \frac{m}{s}$) and medium ($12.21 \frac{m}{s}$) velocity, the pressure distribution is similar, with the medium velocity seeing slightly higher pressure concentrations (darker orange) near the inlet's top and bottom. The high ($48.86 \frac{m}{s}$) velocity pressure distribution contains significantly higher pressure concentrations at the same inlet areas and along the boundary layers of the walls. This indicates that the region near the walls take on significant pressure jumps (both increases and decreases) in the presence of highly turbulent flow, but low to medium velocity changes do not significantly affect the turbulence of any region in the inlet lip.

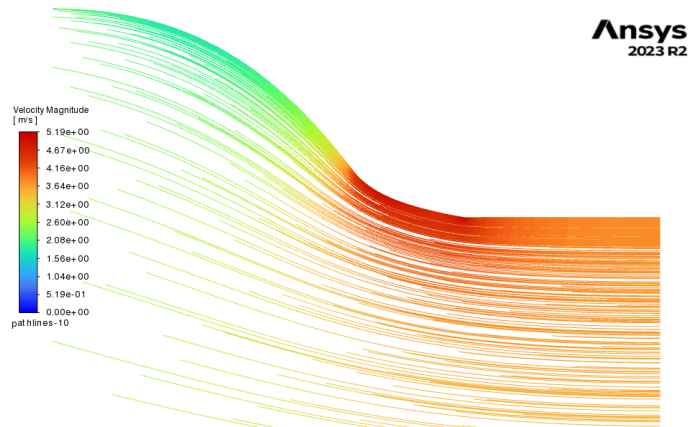


Fig 7. Velocity streamline at $2.44 \frac{m}{s}$ inlet velocity near boundary layer

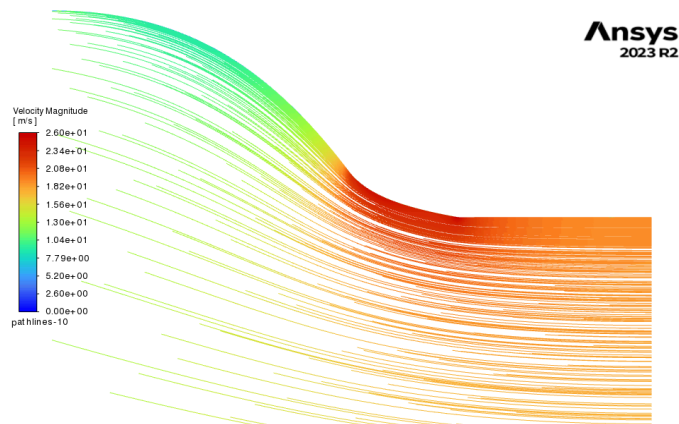


Fig 8. Velocity streamline at $12.21 \frac{m}{s}$ inlet velocity near boundary layer

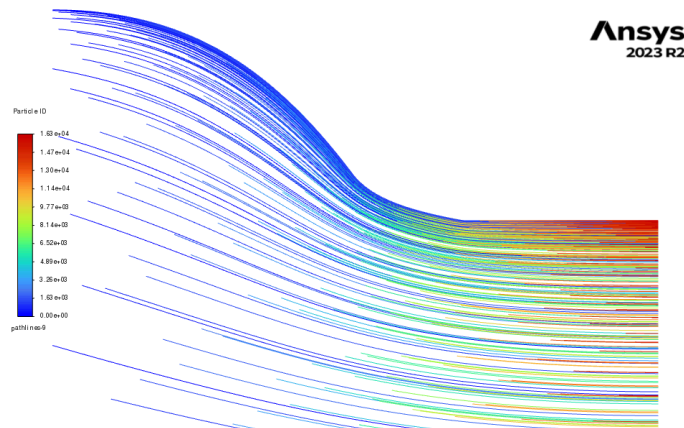


Fig 9. Velocity streamline at $48.86 \frac{m}{s}$ inlet velocity near boundary layer

Similar to the pressure contour plots, the low and medium velocity simulations contain comparable velocity distributions, with the medium turbulent velocity observing a slightly faster flow through the second curve (the red region near the boundary layer) compared to the low velocity one. However, the high turbulence velocity contains a relatively linear increase in velocity towards the end of the inlet, indicating that the flow is accelerating across the entire area (the plot roughly represents half of the symmetrical inlet). The boundary layer is zoomed in to observe this distribution more clearly.

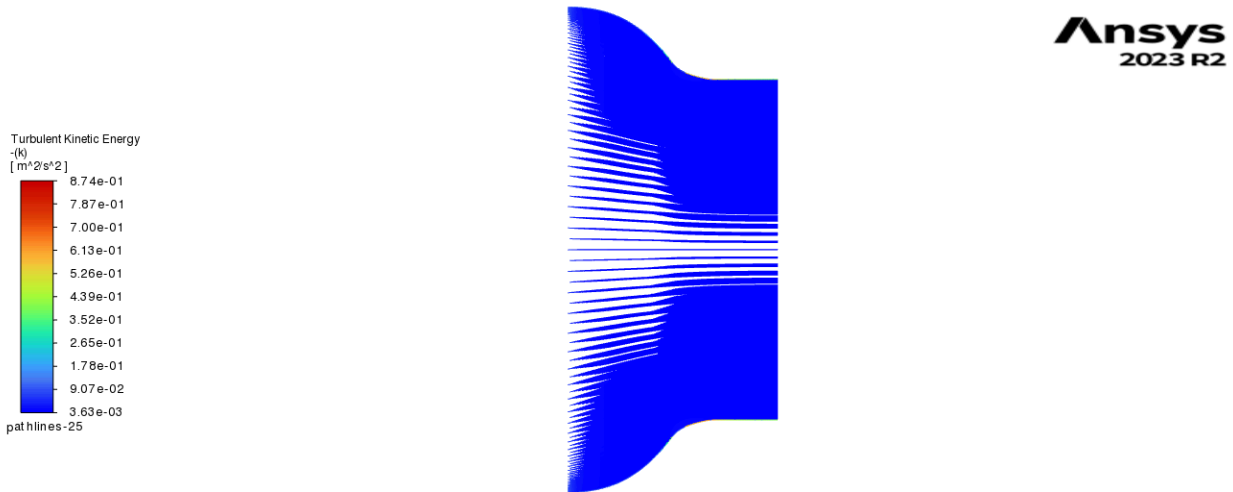


Fig 10. TKE pathline at $2.44 \frac{m}{s}$ inlet velocity

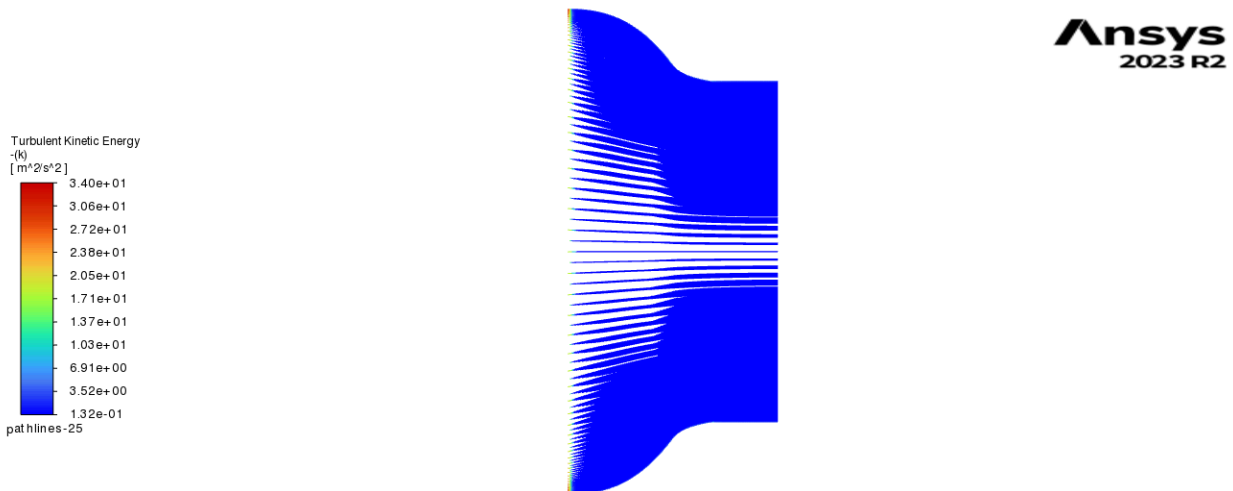


Fig 11. TKE pathline at $12.21 \frac{m}{s}$ inlet velocity

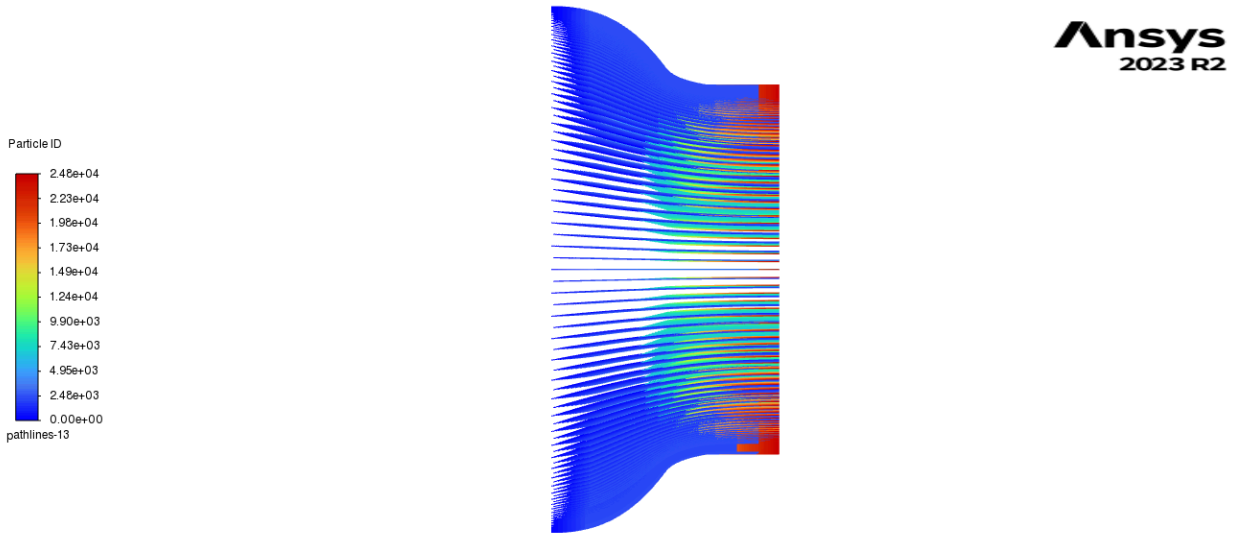


Fig 12. TKE pathline at $48.86 \frac{m}{s}$ inlet velocity

TKE is relatively similar between the low and medium velocity simulations. The medium velocity simulation contains high TKE at the inlet boundary layer, whereas the low velocity simulation contains a consistent low TKE throughout the boundary layers and interior. TKE is significantly less concentrated in the high velocity simulation; TKE starts to increase around the middle of the inlet far from the walls, and jumps even further at the exit-wall areas. Flow out of the exit in the high velocity simulation contains significantly higher TKE than flow into the inlet.

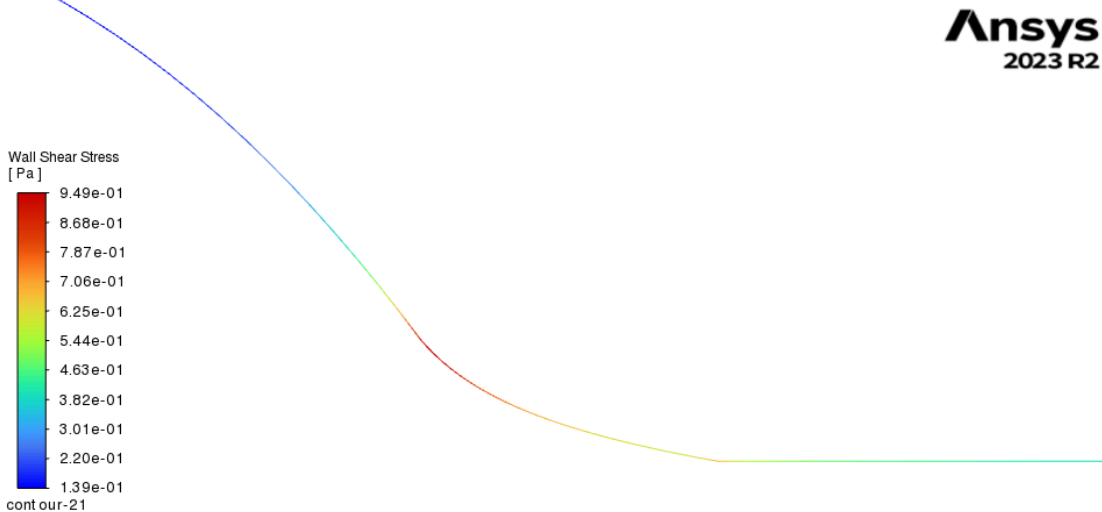


Fig 13. Wall shear stress near boundary layer curve at $2.44 \frac{m}{s}$

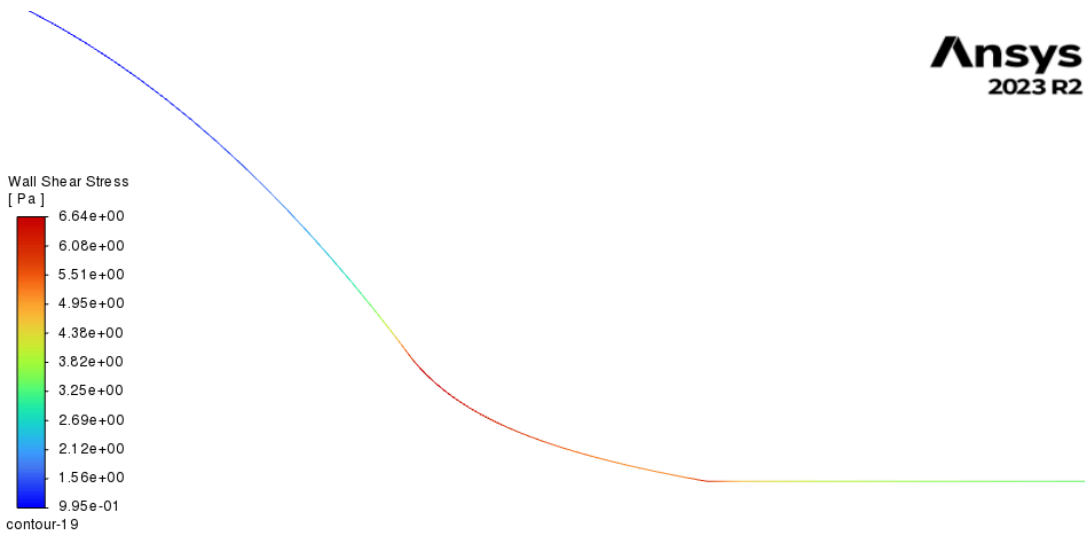


Fig 14. Wall shear stress near boundary layer curve at $12.21 \frac{m}{s}$

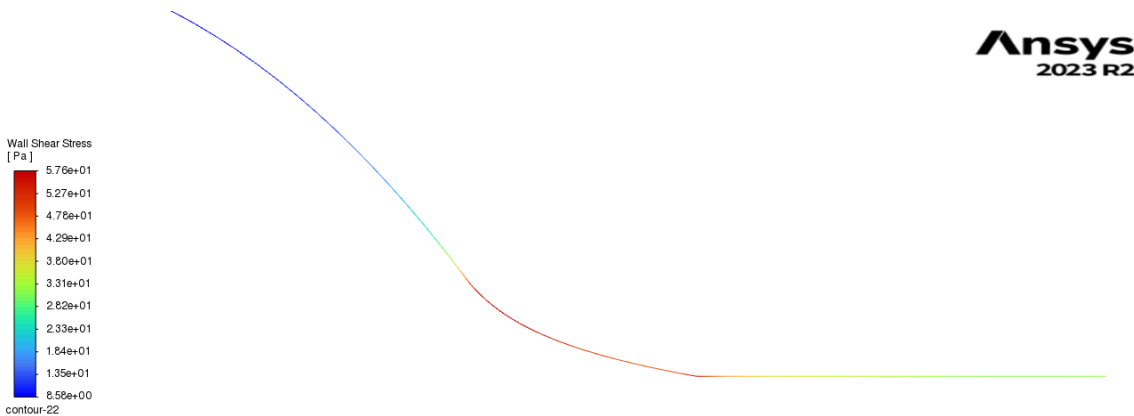


Fig 15. Wall shear stress near boundary layer curve at $48.86 \frac{m}{s}$

Wall shear stress is shown as a pathline of the upper wall. In the low velocity simulation, the upper wall experiences a gradual increase of wall shear stress until the first curve, then gradual decrease to the exit. The medium velocity simulation contains a sharper increase and decrease of wall shear stress at the same points, and the high velocity simulation experiences a similarly distributed wall shear stress with an extended length of maximum shear stress around the entire

first curve. Total wall shear stress is clearly highest on the high velocity simulation, and most evenly distributed in the low velocity simulation. The increased wall shear stress, which leads to quicker flow separation and thinner boundary layers, intuitively corresponds to the increase in TKE of the higher velocity simulation.

Overall, pressure, velocity, and TKE distribution does not have significant change between the $2.44\text{--}12.21 \frac{m}{s}$ range. While the Reynolds number corresponding to velocities above $2.44 \frac{m}{s}$ are turbulent by definition, the geometry of the inlet seems to stabilize the flow such that turbulent parameters, such as TKE and wall shear stress, are tamed. There is a drastic change for all of these flow parameters, including wall shear stress, somewhere along the $12.21\text{--}48.86 \frac{m}{s}$ range. Many performance characteristics seem to be favorable in the high velocity simulation despite the extreme turbulence; static pressure is increased at the inlet with little effect on downstream pressure losses, and flow is gradually accelerated and enters the fan section at maximum speed. The chaos of turbulence visually increases significantly with the high velocity simulation, but its correlation with fan performance cannot be broadly concluded as boundary layer mixing, flow geometry, accompanying pressure losses, and blade loading and stall all affect fan performance in either direction.

4. Conclusion

In fan applications, turbulence is not inherently detrimental to performance. Flow velocity direction remains largely consistent across low- and high-turbulence simulations, as the subsonic forward velocity and inlet geometry prevent significant swirling or eddies, which are unfavorable for fan performance. Velocity distribution is most efficient in the high-turbulence simulation, where well-mixed, high-speed flow is ideal for air exiting the inlet and entering the fan section. Static pressure distribution remains largely unchanged, suggesting turbulence location is unaffected by the magnitude of pressure.

Turbulent kinetic energy (TKE) and shear stress increase notably in the high-turbulence simulation while remaining stable in low- and medium-turbulence scenarios. However, the relationship between these turbulent parameters and fan performance is inconclusive and warrants further investigation, including static and dynamic thrust testing and more detailed CFD analysis across a broader range of turbulence conditions. Based on velocity distribution, this inlet geometry appears optimized for high-speed, high-turbulence scenarios, assuming turbulence at the exit does not induce significant pressure losses or flow separation.

5. Reflection Section

Through this report, what stuck out to me was the sheer amount of variables that affect what I saw in the simulations, from the aerodynamic considerations in Fluent to grid design to boundary

conditions to solver settings. It reinforced how robust a fundamental understanding of science (which was brought up in the beginning of the course) is required to run useful simulations. It also showed how integrity in every step is crucial, especially when designing meshes, running simulations, and visualizing results are done in different software. Everything stems from the precision of mesh design, which draws from the math we spent a lot of time developing over the semester, and I saw that come together for this project. Throughout writing this report, I also understand a little more how simulation results are worthless without proper written communication, as CFD was a lot more open-ended in investigation than I anticipated.

References

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